

CAD Design Audit: drive-unit

Mechanical design review

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1. Model Overview

Source file **drive-unit.step** contains 48 entities, 8 distinct products and 5 solid bodies. Total placed instances: 17.

Part	Quantity
608ZZ Ball Bearing	2
Chassis Plate	1
M2x6 Pan Screw	4
M3x12 SHCS	2
M3x8 SHCS	4
Motor Mount	2
N20 Gearmotor	2

[Insert figure here]

Figure 1: Assembly overview of drive-unit

2. Fastener Inventory

10 fasteners across 3 sizes.

Size	Quantity	Part names
M2 x 6	4	M2x6 Pan Screw
M3 x 12	2	M3x12 SHCS
M3 x 8	4	M3x8 SHCS

3. Bearing Inventory

2 bearings across 1 size.

Size	Quantity	Part names
608ZZ	2	608ZZ Ball Bearing

4. Other Components

Part	Quantity
Chassis Plate	1
Motor Mount	2

Part	Quantity
N20 Gearmotor	2

5. Load and Design Concerns

Reviewed item by item during the audit. Each point below is answered in the findings section.

5.1 Fasteners

Check thread engagement depth against 1.5 x nominal diameter in plastic and 1.0 x in metal. Check edge distance at every tapped or clearance hole. Confirm no fastener carries shear it is not sized for.

5.2 Bearings

Check seat tolerance and fit class for each bearing bore and shaft. Confirm axial retention on both races. Flag any bearing loaded outside its rated direction.

5.3 Load Paths

Trace the primary load path from input to ground. Flag cantilevered members, unsupported spans, and single-fastener moment joints.

6. Findings and Recommendations

Completed by the reviewing engineer for the delivered audit.

1. Finding: (state the issue, the location, and the severity) 2. Recommendation: (state the change and why it resolves the finding)